NOBEL PRIZE IN MEDICINE 2026

What If Light Could Unlock the Most Complex Organ in the Human Body?!

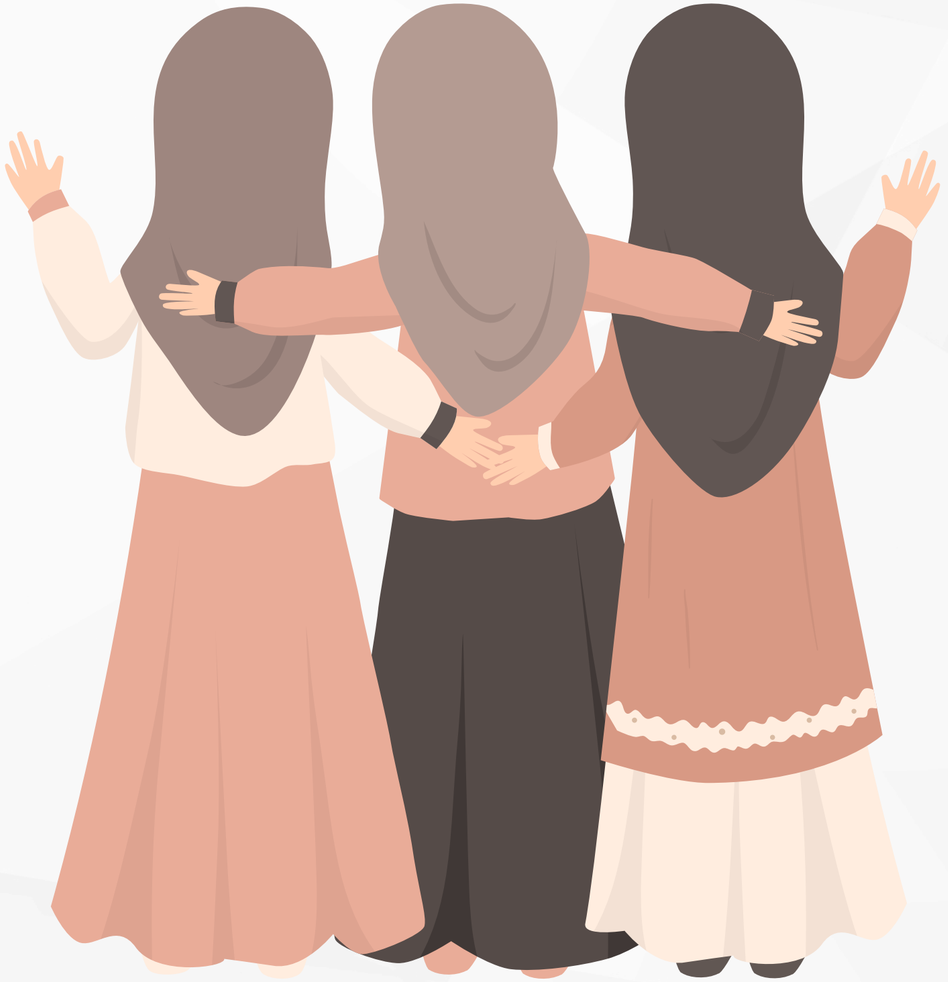


Presented by Group 1
Under the supervision of Prof. Magdy Ghannam

Members of the group







Scan the ID to get yours

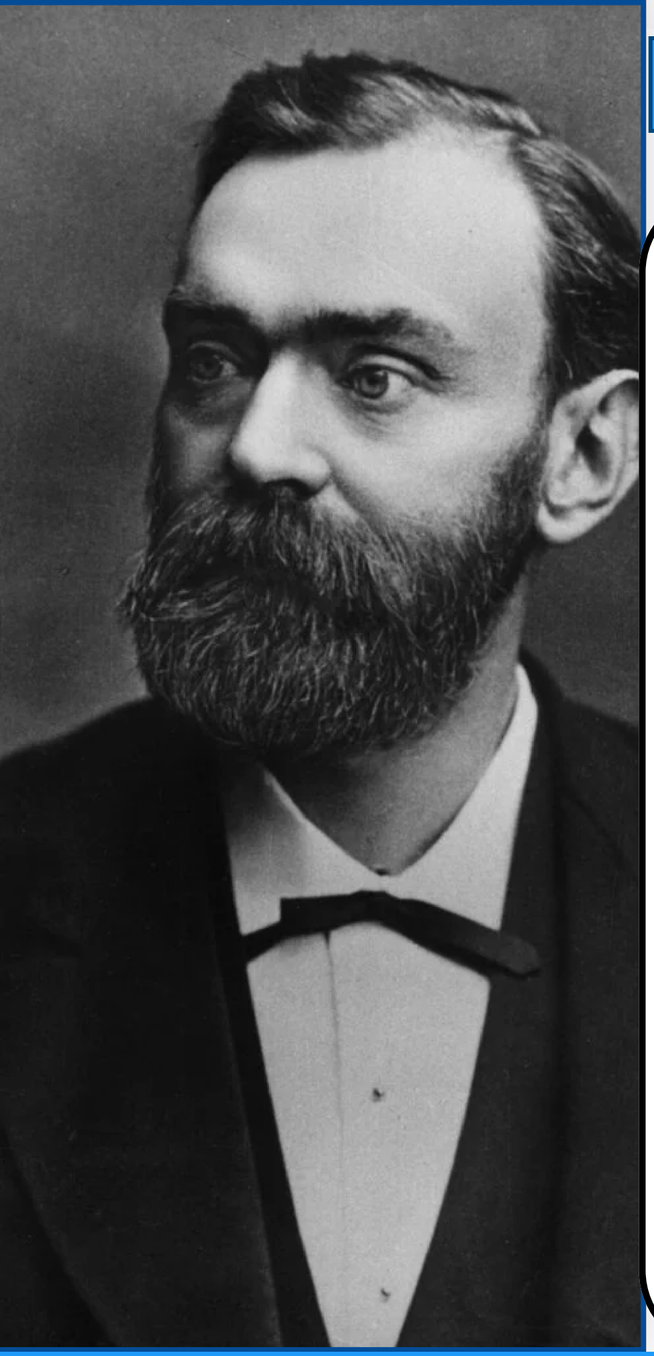
Saleh Sayed

Nourhan Magdy

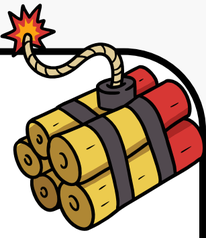


Rehab Zaher

Maha Hossam



HISTORY BEHIND ALFRED NOBEL



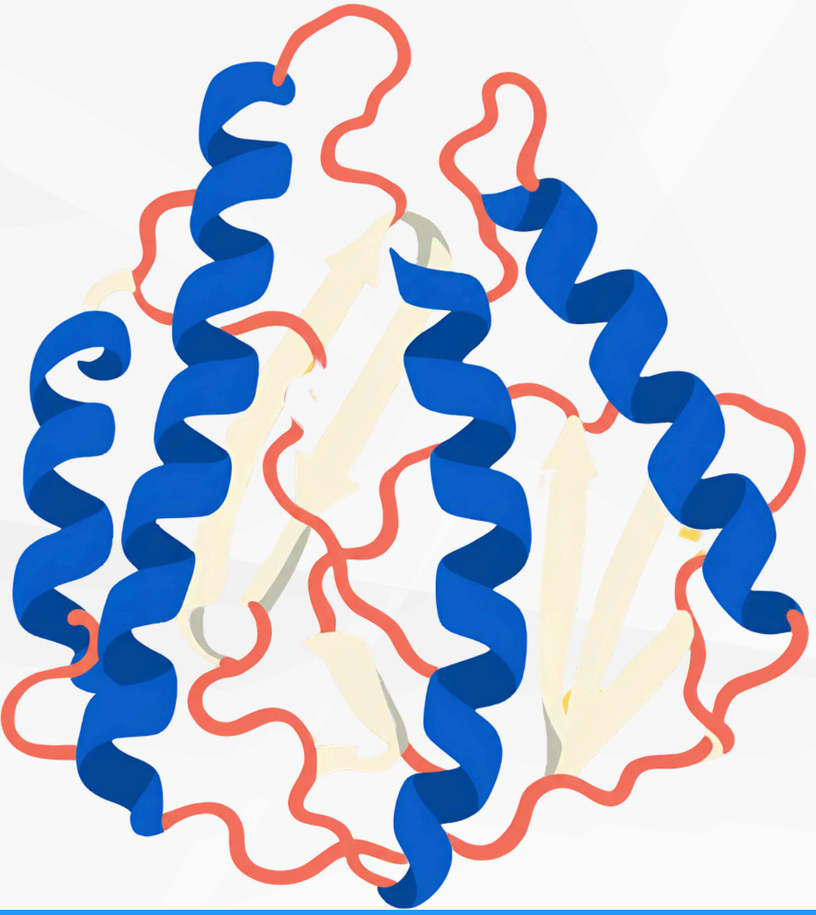
- ALFRED NOBEL WAS A SWEDISH SCIENTIST BORN IN 1833, PASSIONATE ABOUT SCIENCE, ESPECIALLY CHEMISTRY.
- IN 1867, HE INVENTED DYNAMITE, WHICH LATER BECAME ASSOCIATED WITH DESTRUCTION.
- A YEAR BEFORE HIS DEATH, HE WROTE A WILL ESTABLISHING AWARDS TO HONOR CONTRIBUTIONS TO HUMANITY.
- THE NOBEL PRIZES RECOGNIZE ACHIEVEMENTS IN PHYSICS, CHEMISTRY, PHYSIOLOGY OR MEDICINE, LITERATURE, AND PEACE. ECONOMIC SCIENCES WAS ADDED LATER.
- NOBEL DIED ON DECEMBER 10, 1896, LEAVING BEHIND A LEGACY THAT CELEBRATES HUMAN ACHIEVEMENT EVERY YEAR.

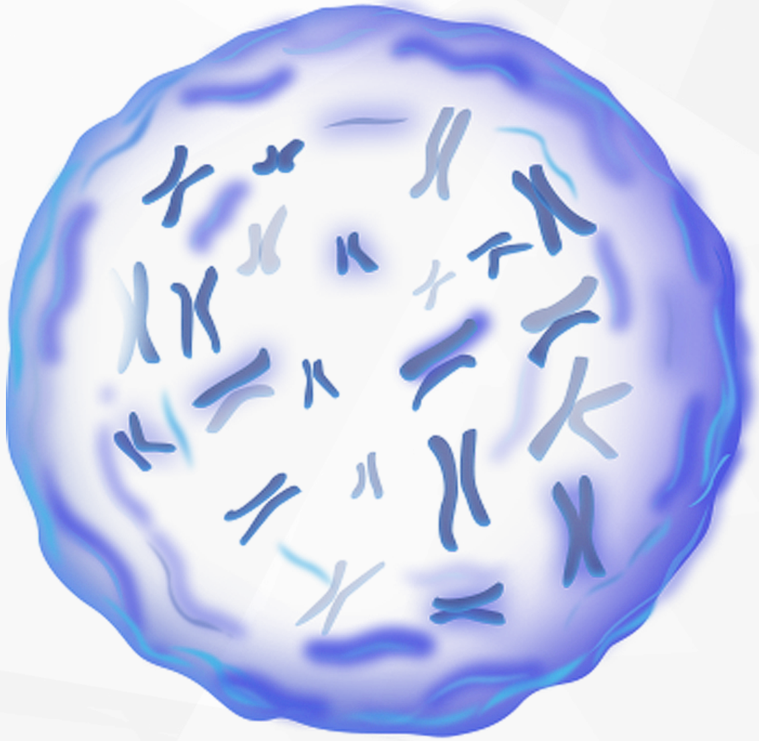


BREAKING 🚨: The 2026 Nobel Prize in Physiology or Medicine has been awarded to Karl Deisseroth, Peter Hegemann and Georg Nagel for their discoveries of light-gated ion channels and optogenetics — a method that allows scientists to control nerve cells using light.

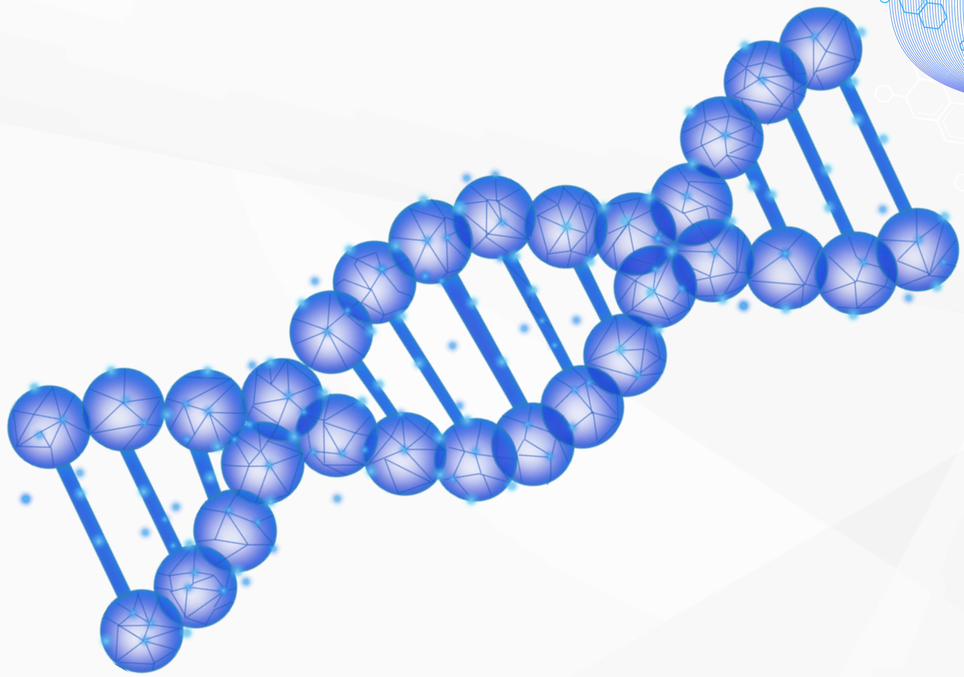
BREAKING NEWS

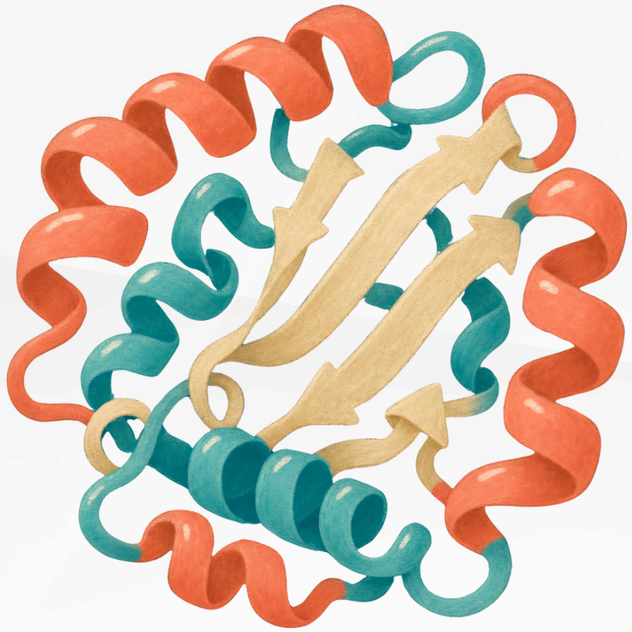
The Scientific Fields Behind Optogenetics



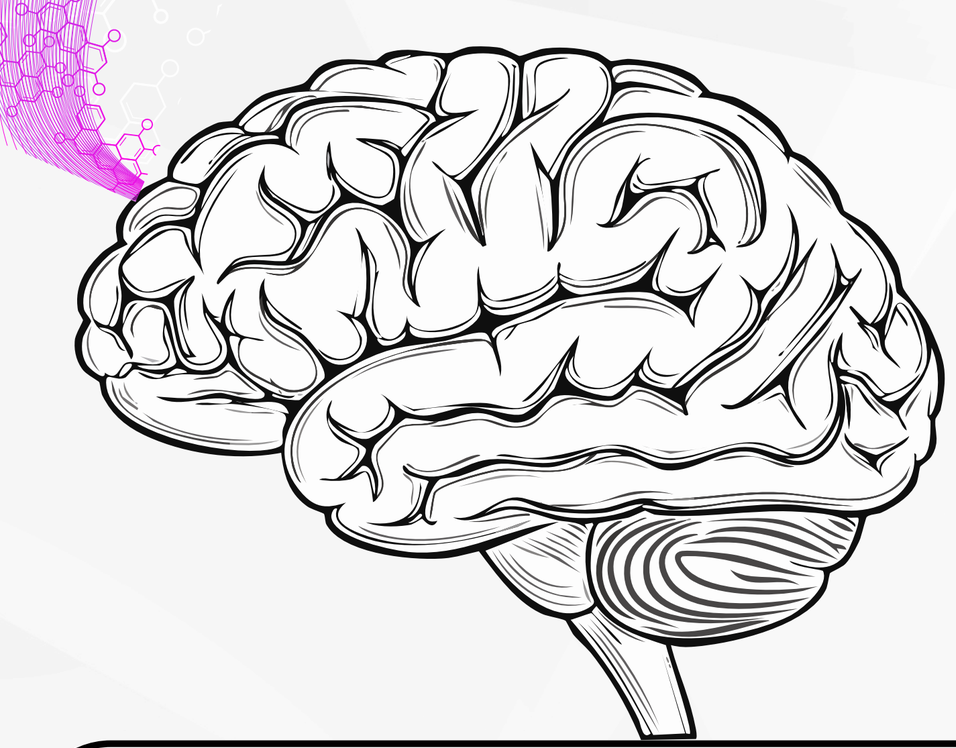


BIOPHYSICS HELPED EXPLAIN HOW LIGHT ACTIVATES CHANNELRHODOPSIN-2 (CHR2), A LIGHT-SENSITIVE PROTEIN, AND HOW THIS ACTIVATION ALLOWS IONS TO CROSS THE CELL MEMBRANE. UNDERSTANDING THESE ELECTRICAL AND PHYSICAL CHANGES MADE IT POSSIBLE TO CONTROL NEURONAL ACTIVITY USING LIGHT.



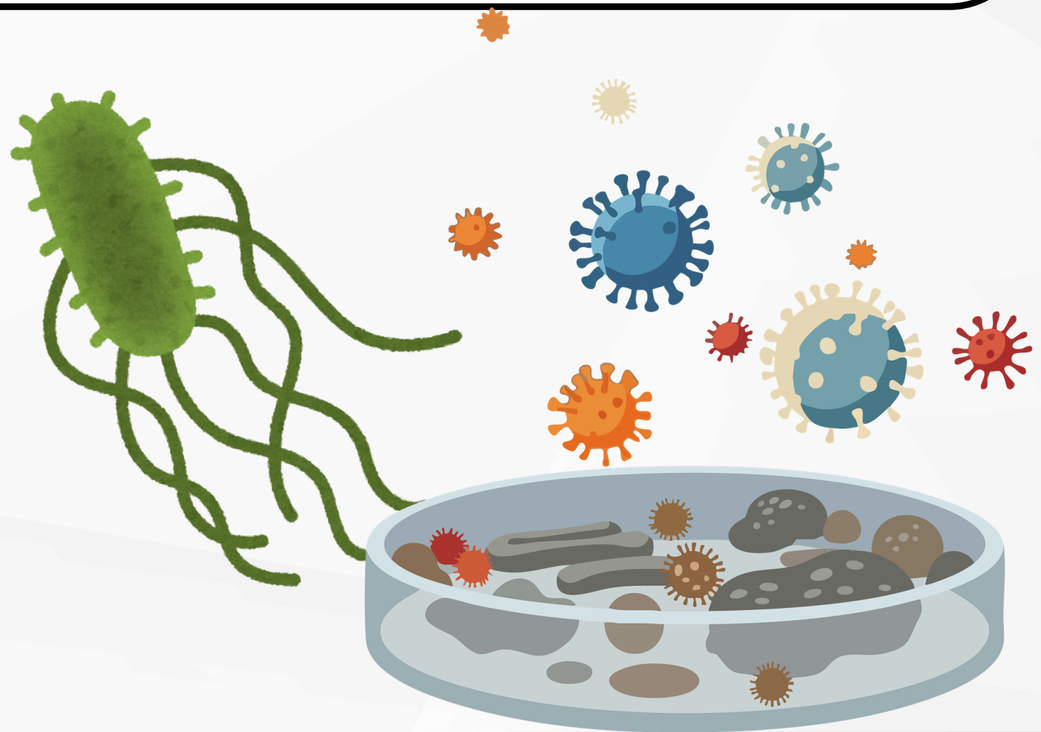


MOLECULAR BIOLOGY AND GENETICS HELPED SCIENTISTS UNDERSTAND THE GENES THAT ENCODE LIGHT-SENSITIVE PROTEINS. BY INTRODUCING THE GENE FOR CHR2 INTO NEURONS, RESEARCHERS ENABLED THESE CELLS TO RESPOND TO LIGHT, CREATING THE FOUNDATION FOR OPTOGENETIC CONTROL.



MICROBIOLOGY AND PHYCOLOGY CONTRIBUTED TO THE DISCOVERY OF LIGHT-SENSITIVE PROTEINS IN MICROORGANISMS AND ALGAE, PARTICULARLY CHLAMYDOMONAS REINHARDTII. STUDYING THESE PROTEINS, INCLUDING CHANNELRHODOPSINS, INSPIRED SCIENTISTS TO DEVELOP OPTOGENETICS AS A TOOL FOR CONTROLLING THE ACTIVITY OF NEURONS USING LIGHT.

NEUROSCIENCE PROVIDED THE KNOWLEDGE NEEDED TO UNDERSTAND HOW NEURONS COMMUNICATE AND FORM CIRCUITS THAT CONTROL BRAIN FUNCTIONS. OPTOGENETICS ALLOWS RESEARCHERS TO ACTIVATE OR INHIBIT SPECIFIC NEURONS WITH LIGHT, HELPING THEM INVESTIGATE HOW THESE CIRCUITS CONTRIBUTE TO BEHAVIOR AND BRAIN ACTIVITY.





Research Gap

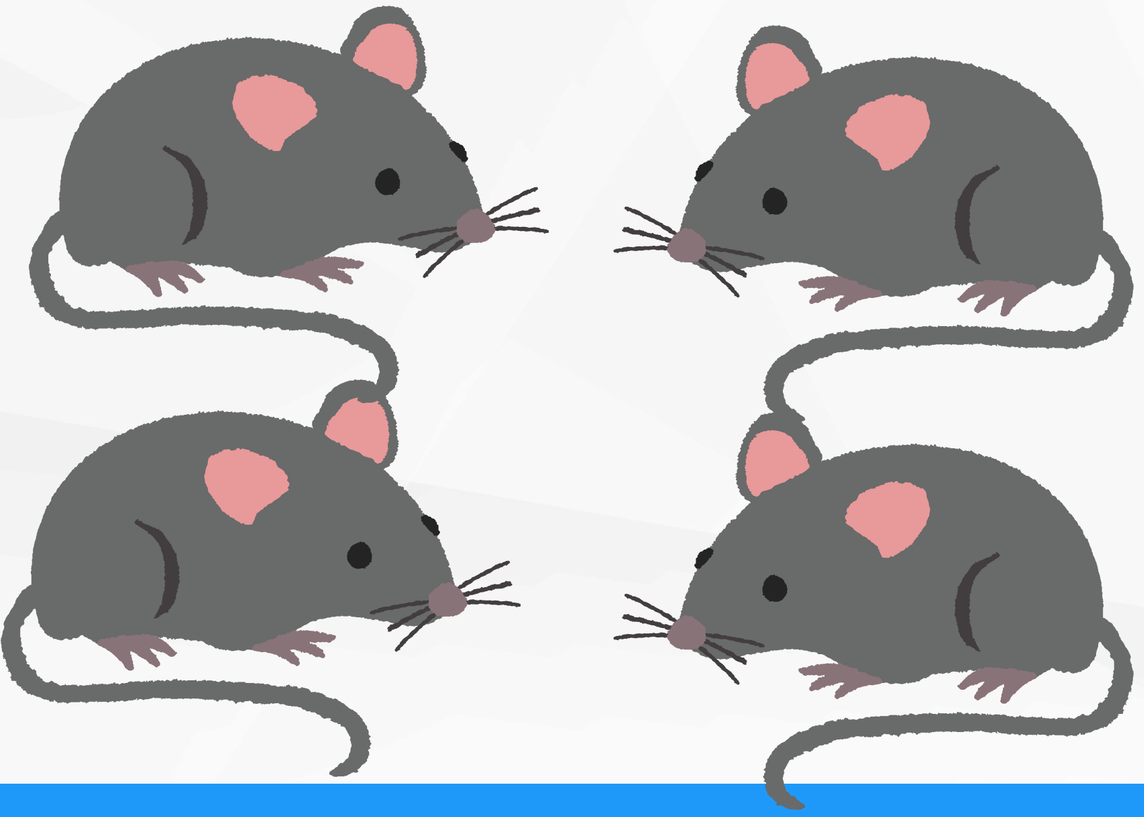


Search

Can a specific type of neuron be directly linked to a particular behavioral response



Beyond Correlation

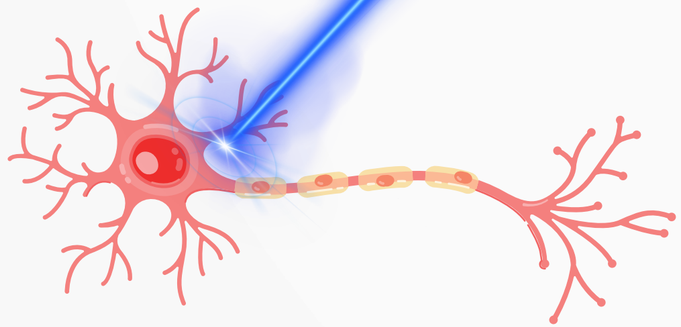


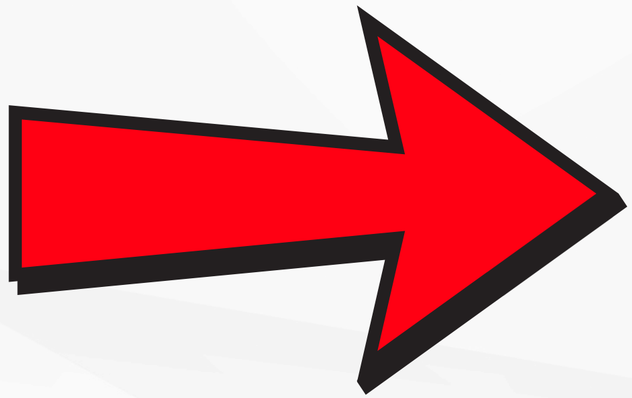


Proving Causation

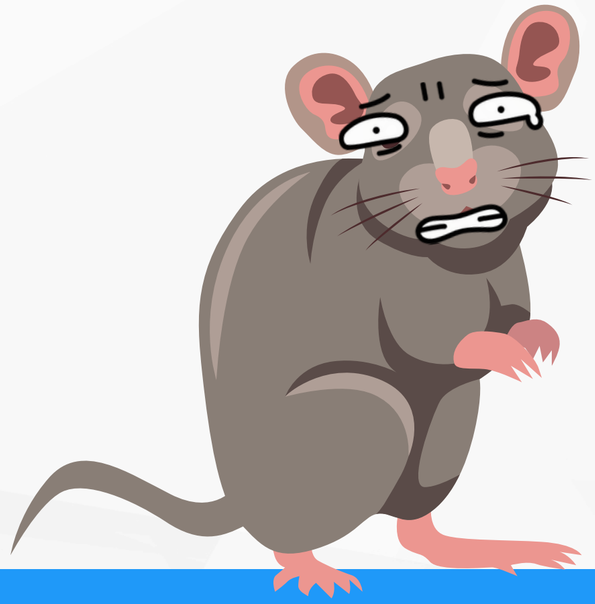
Safety response

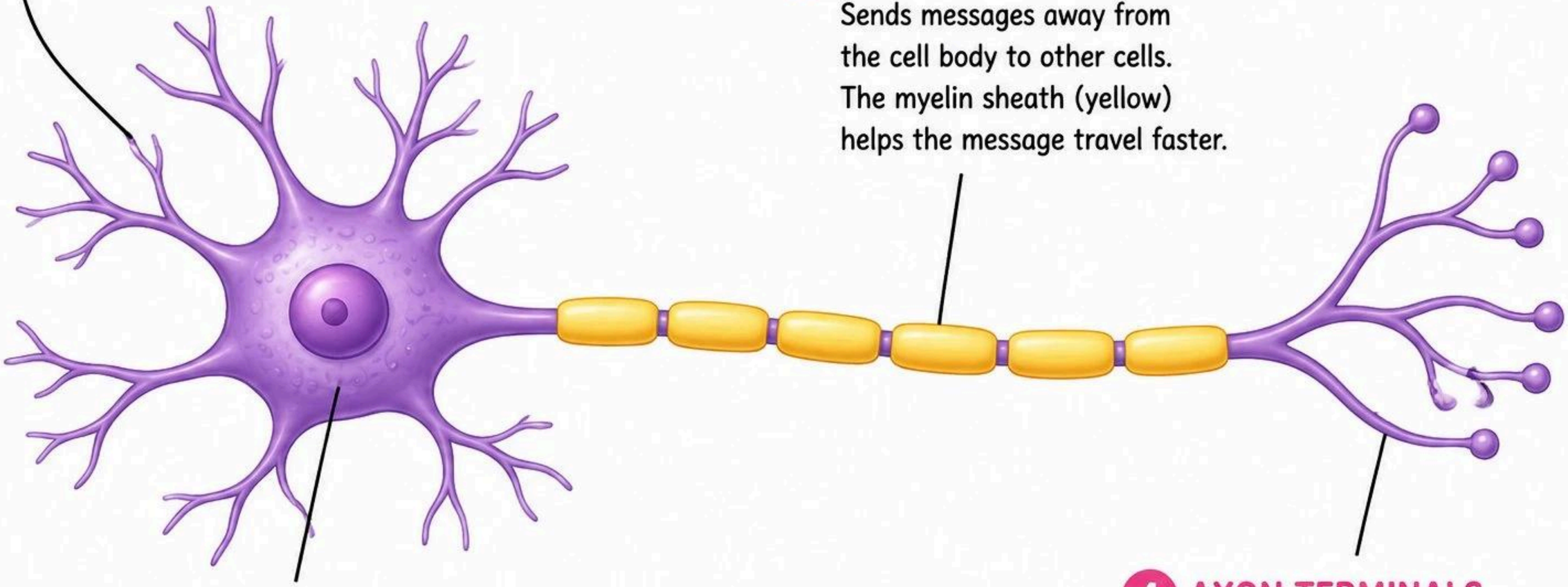






Fear response





1 DENDRITES

Receive messages from other cells and carry them toward the cell body.

2 CELL BODY (SOMA)

The control center of the neuron. It contains the nucleus and keeps the cell alive.

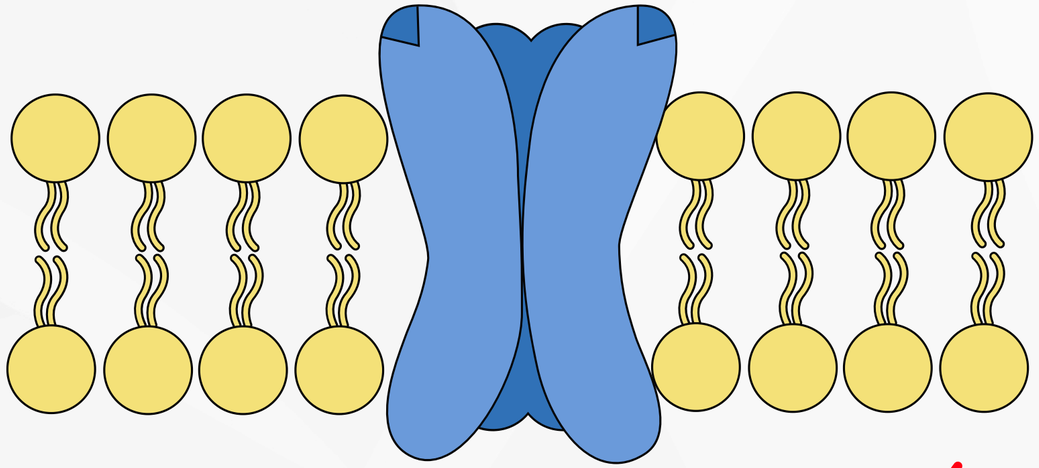
3 AXON

Sends messages away from the cell body to other cells. The myelin sheath (yellow) helps the message travel faster.

4 AXON TERMINALS

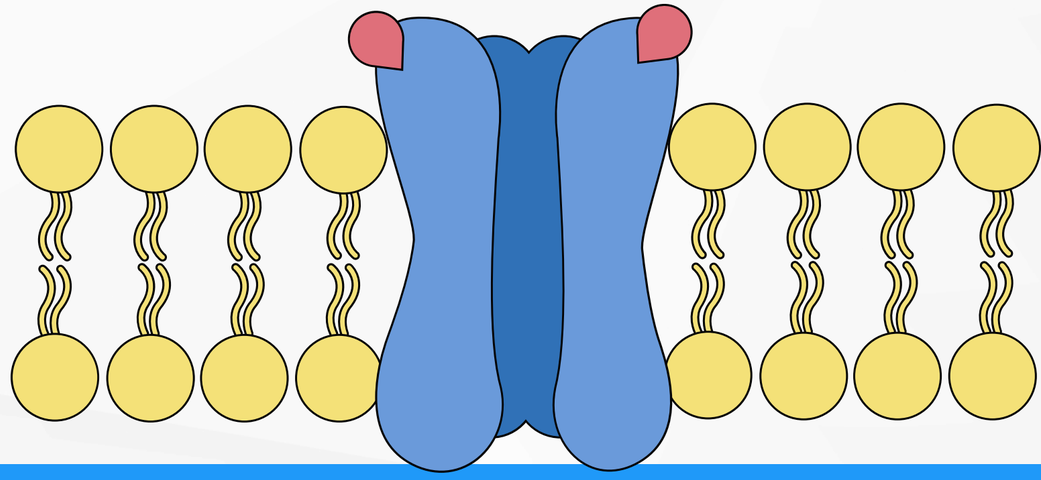
The ends of the axon. They send messages to the next cell across a small gap called a synapse.

Channelrhodopsin-2

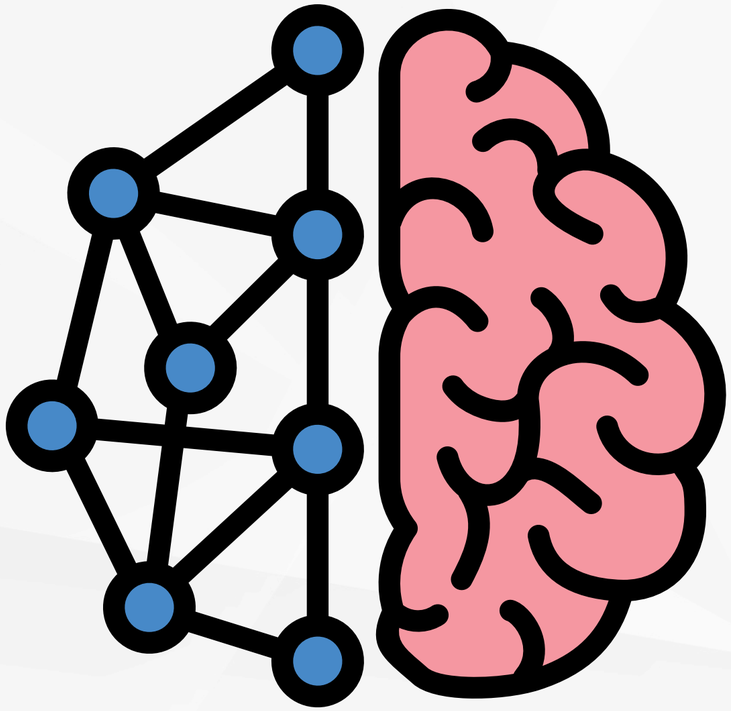


Closed

Opened

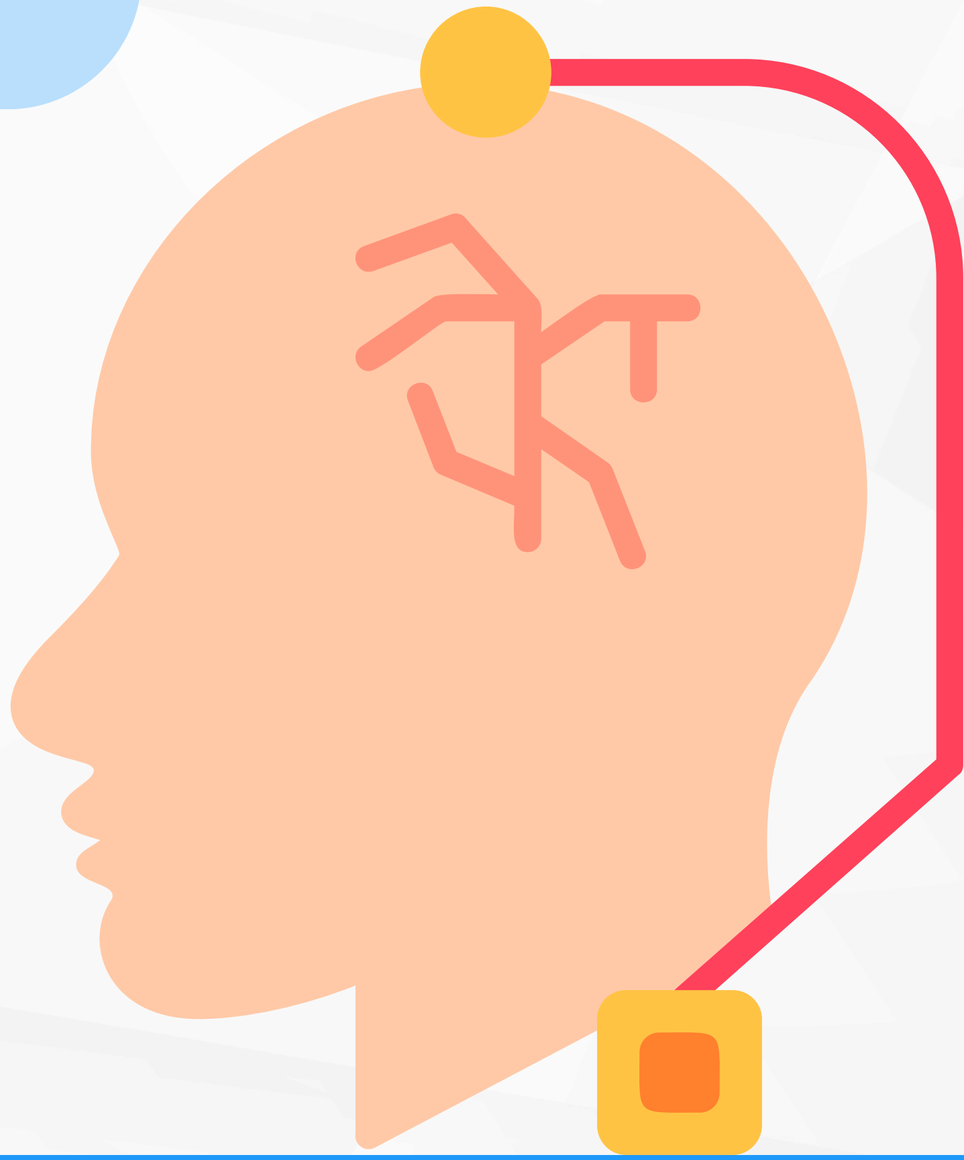


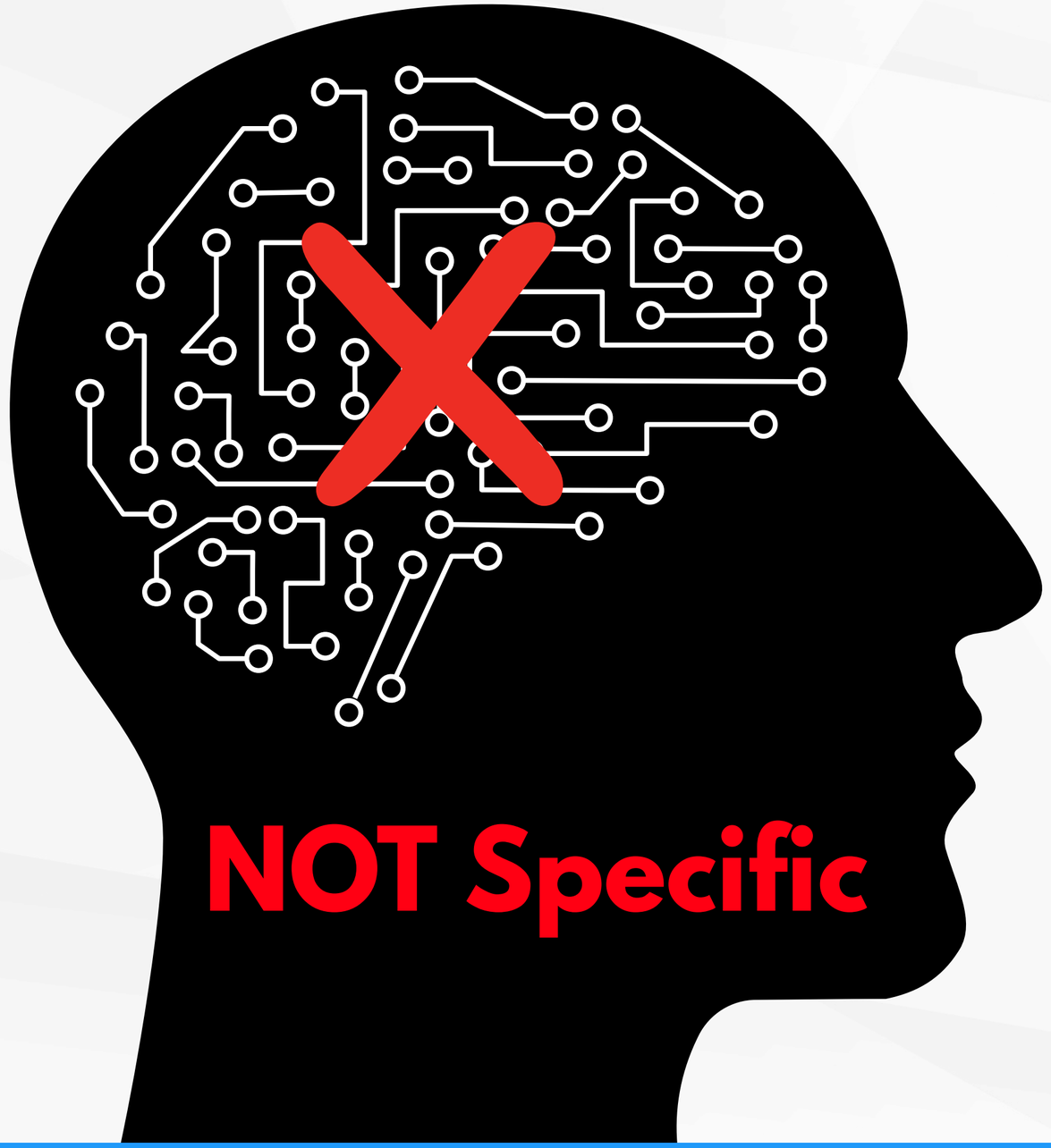
Traditional Neuromodulation Techniques



Neural Circuits

DBS

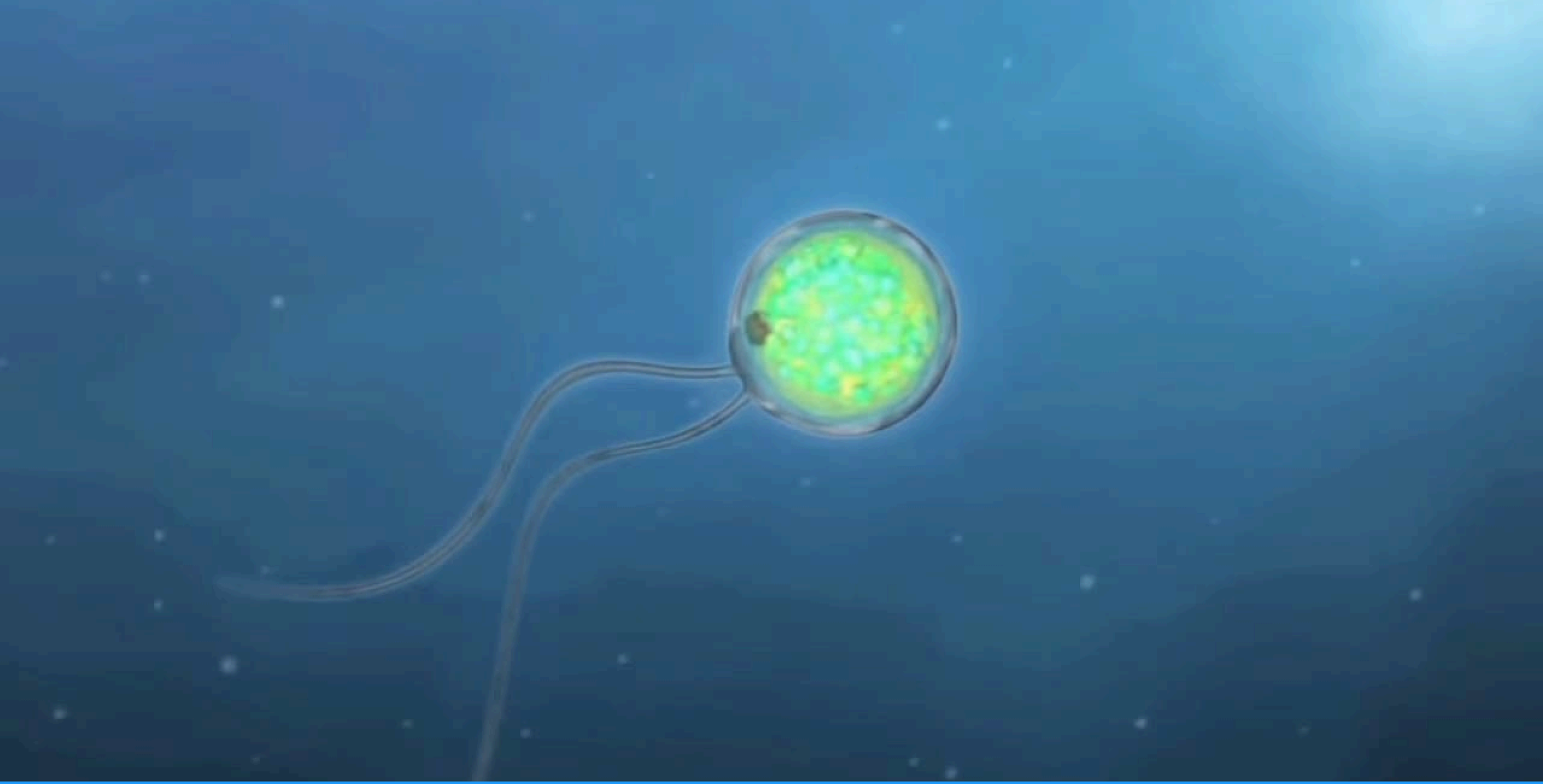



NOT Specific


CONS

SOLUTION





Think carefully to get your gift!!

